

Constructors

A constructor is a **special method** that is automatically called when an object of a class is created. To create a constructor, use the same name as the class, followed by parentheses ():

Example

```
#include <iostream>
using namespace std;
class MyClass {    // The class
public:            // Access specifier
    MyClass() {    // Constructor
        cout << "Hello World!";
    }
};

int main() {
    MyClass myObj;    // Create an object of MyClass (this will call the constructor)
    return 0;
}
```

Output:
Hello World!

Constructor with Parameters

Example:

```
#include <iostream>
using namespace std;
class Car {
public:            // Access specifier
    string brand; // Attribute
    string model; // Attribute
    int year;     // Attribute
    Car(string x, string y, int z) { // Constructor with parameters
        brand = x;
        model = y;
        year = z;
    }
};

int main() {
    // Create Car objects and call the constructor with different values
    Car carObj1("BMW", "X5", 2019);
    Car carObj2("Ford", "Mustang", 2021);

    // Print values
    cout << carObj1.brand << " " << carObj1.model << " " << carObj1.year << "\n";
    cout << carObj2.brand << " " << carObj2.model << " " << carObj2.year << "\n";
    return 0;
}
```

Output:
BMW X5 2019
Ford Mustang 2021

Constructor Defined Outside the Class

You can also define the constructor outside the class using the scope resolution operator `::`

Example

```
class Car {           // The class
public:              // Access specifier
    string brand;    // Attribute
    string model;    // Attribute
    int year;        // Attribute
    Car(string x, string y, int z); // Constructor declaration
};

// Constructor definition outside the class
Car::Car(string x, string y, int z) {
    brand = x;
    model = y;
    year = z;
}

int main() {
    // Create Car objects and call the constructor with different values
    Car carObj1("BMW", "X5", 2019);
    Car carObj2("Ford", "Mustang", 2021);

    // Print values
    cout << carObj1.brand << " " << carObj1.model << " " << carObj1.year << "\n";
    cout << carObj2.brand << " " << carObj2.model << " " << carObj2.year << "\n";
    return 0;
}
```

Output:
BMW X5 2019
Ford Mustang 2021